

Tuesday Warm-up: Lorentz Force Law, and Biot-Savart Law

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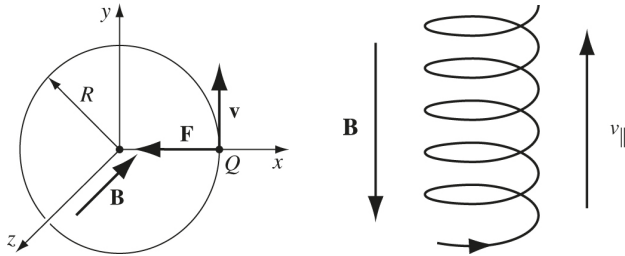


Figure 1: (Left) Uniform circular motion of a charge Q within a magnetic field B . (Right) Helical motion within a field, with average velocity $v_{\parallel}$.

1 Memory Bank

- **Lorentz force:**

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (1)$$

- **Position for uniform circular motion (xy -plane):**

$$\mathbf{r}(t) = |\mathbf{r}| \cos(\omega t) \hat{i} + |\mathbf{r}| \sin(\omega t) \hat{j} \quad (2)$$

- **Centripetal force:**

$$\mathbf{F}_C = -mr\omega^2 \hat{\mathbf{r}} \quad (3)$$

$$\mathbf{F}_C = -mv^2 r^{-1} \hat{\mathbf{r}} \quad (4)$$

- **Mass and charge of proton:**

$$q = 1.602 \times 10^{-19} \text{ C} \quad (5)$$

$$m = 1.673 \times 10^{-27} \text{ kg} \quad (6)$$

2 Lorentz Force

1. Consider Fig. 1 (left). (a) Show that, if $\mathbf{E} = 0$, the motion is circular, with constant frequency

$$\omega_C = \frac{qB}{m} \quad (7)$$

The parameter ω_C is known as the *cyclotron frequency*. (b) What is ω_C for a proton circling in a 1 Tesla magnetic field? (c) Suppose $\mathbf{r}(0) = 1.0\hat{i}$ m. Where is the proton at $t = 7.83$ ns? (d) If the proton moves along the z -axis with $v_z = 0.003 \text{ m ns}^{-1}$, what is the position at 2.61 ns? The motion is depicted in Fig. 1 (right).

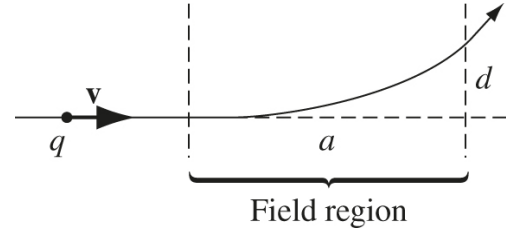


Figure 2: A charge passes through a region with a $\mathbf{B}$ -field *into* the page.

2. A particle of charge q enters a region of uniform field $\mathbf{B}$ (pointing *into* the page). The field deflects the particle a distance d above the original line of flight (Fig. 2). (a) Is the charge negative or positive? (b) In terms of a , d , B , and q , find the momentum of the particle.

3. In 1897, J.J. Thomson first measured the charge-to-mass ratio of electrons.

- First he passed the beam of cathode rays through uniform crossed $\mathbf{E}$ and $\mathbf{B}$ fields. These were mutually perpendicular, and both perpendicular to the beam. He adjusted the $\mathbf{E}$ field until he got zero deflection. What was the speed of the electrons, in terms of $\mathbf{E}$ and $\mathbf{B}$?
- Then he turned off $\mathbf{E}$ and measured the radius of curvature R of the beam. In terms of E , B , and R , what is (q/m) for the beam?